

Complex Numbers

ANSWER KEY



Arithmetic with complex numbers

- 1 Let $z_1 = 4 - 2i$ and $z_2 = -3 + i$. Find $z_1 + z_2$ and $z_1 z_2$.

$$z_1 + z_2 = (4 - 3) + (-2 + 1)i = 1 - i. \quad z_1 z_2 = (4)(-3) - (-2)(1) + [(4)(1) + (-2)(-3)]i = -10 + 10i.$$

- 2 Find $(6 + i) - (2 - 5i)$.

$$= 4 + 6i$$

Complex roots of a quadratic

- 3 Solve $x^2 + 6x + 13 = 0$.

$$\Delta = 6^2 - 4(1)(13) = 36 - 52 = -16 < 0. \text{ By Theorem 1.1, } x = \frac{-6 \pm \sqrt{-16}}{2} = \frac{-6 \pm 4i}{2} = -3 \pm 2i.$$

- 4 Solve $2x^2 - 4x + 10 = 0$.

$$\Delta = (-4)^2 - 4(2)(10) = 16 - 80 = -64 < 0. \quad x = \frac{4 \pm \sqrt{-64}}{4} = \frac{4 \pm 8i}{4} = 1 \pm 2i.$$

- 5 Without solving it, explain how you know $x^2 + 2x + 5 = 0$ has complex (not real) roots.

Its discriminant is $\Delta = 2^2 - 4(1)(5) = 4 - 20 = -16 < 0$. A negative discriminant means the quadratic formula takes the square root of a negative number, giving a complex conjugate pair (Theorem 1.1) instead of real roots.

- 6 Verify that $x = -3 + 2i$ is a root of $x^2 + 6x + 13 = 0$ by direct substitution.

$$(-3 + 2i)^2 = 9 - 12i + 4i^2 = 9 - 12i - 4 = 5 - 12i. \text{ And } 6(-3 + 2i) = -18 + 12i. \text{ Adding: } (5 - 12i) + (-18 + 12i) + 13 = 0. \text{ Yes, it checks out.}$$

Modulus

- 7 Find $|z|$ for $z = -3 + 2i$ (one of the roots found above).

$$|z| = \sqrt{(-3)^2 + 2^2} = \sqrt{13}.$$

- 8 Find $|z|$ for $z = 5 - 12i$.

$$|z| = \sqrt{5^2 + 12^2} = \sqrt{169} = 13.$$

- 9 Explain why a conjugate pair $a \pm bi$ always has the same modulus for both roots.

$|a + bi| = \sqrt{a^2 + b^2}$ and $|a - bi| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2}$ — squaring b or $-b$ gives the same result, so the two moduli are identical.

Multiple choice — easy

- 10 Let $z_1 = -6 + 9i$ and $z_2 = 2 + 6i$. Find $z_1 + z_2$.

- a. $-2 + 15i$
- b. $-4 + 13i$
- c. $-4 + 15i$ ✓
- d. $-8 + 3i$

$$z_1 + z_2 = (-6 + 2) + (9 + 6)i = -4 + 15i.$$

- 11 Let $z_1 = -2 - 3i$ and $z_2 = -4 - 8i$. Find $z_1 + z_2$.

- a. $-6 - 13i$
- b. $-6 - 11i$ ✓
- c. $2 + 5i$
- d. $-4 - 11i$

$$z_1 + z_2 = (-2 - 4) + (-3 - 8)i = -6 - 11i.$$

- 12 Let $z_1 = 3 + 8i$ and $z_2 = -9 - 7i$. Find $z_1 + z_2$.

- a. $12 + 15i$
- b. $-6 - i$
- c. $-4 + i$
- d. $-6 + i$ ✓

$$z_1 + z_2 = (3 - 9) + (8 - 7)i = -6 + i.$$

13 Let $z_1 = 7 + 9i$ and $z_2 = -5 + 5i$. Find $z_1 + z_2$.

a. $2 + 12i$

b. $2 + 14i$ ✓

c. $12 + 4i$

d. $4 + 14i$

$$z_1 + z_2 = (7 - 5) + (9 + 5)i = 2 + 14i.$$

14 Let $z_1 = -8 - 8i$ and $z_2 = 5 + 3i$. Find $z_1 + z_2$.

a. $-13 - 11i$

b. $-3 - 5i$ ✓

c. $-1 - 5i$

d. $-3 - 7i$

$$z_1 + z_2 = (-8 + 5) + (-8 + 3)i = -3 - 5i.$$

15 Let $z_1 = 8 + 5i$ and $z_2 = 9 - 3i$. Find $z_1 + z_2$.

a. $19 + 2i$

b. $17 + 2i$ ✓

c. 17

d. $-1 + 8i$

$$z_1 + z_2 = (8 + 9) + (5 - 3)i = 17 + 2i.$$

16 Let $z_1 = -7 - 3i$ and $z_2 = -4 + 2i$. Find $z_1 + z_2$.

a. $-3 - 5i$

b. $-11 - 3i$

c. $-9 - i$

d. $-11 - i$ ✓

$$z_1 + z_2 = (-7 - 4) + (-3 + 2)i = -11 - i.$$

17 Let $z_1 = 3 - 7i$ and $z_2 = -1 - i$. Find $z_1 + z_2$.

a. $2 - 8i$ ✓

b. $4 - 6i$

c. $4 - 8i$

d. $2 - 10i$

$$z_1 + z_2 = (3 - 1) + (-7 - 1)i = 2 - 8i.$$

18 Let $z_1 = 1 + 8i$ and $z_2 = -2 - 4i$. Find $z_1 + z_2$.

a. $3 + 12i$

b. $1 + 4i$

c. $-1 + 2i$

d. $-1 + 4i$ ✓

$$z_1 + z_2 = (1 - 2) + (8 - 4)i = -1 + 4i.$$

19 Let $z_1 = 5 - i$ and $z_2 = -3 - 6i$. Find $z_1 + z_2$.

a. $2 - 9i$

b. $8 + 5i$

c. $2 - 7i$ ✓

d. $4 - 7i$

$$z_1 + z_2 = (5 - 3) + (-1 - 6)i = 2 - 7i.$$

20 Let $z_1 = 3 + 7i$ and $z_2 = -9 + 4i$. Find $z_1 + z_2$.

- a. $-4 + 11i$
- b. $-6 + 9i$
- c. $-6 + 11i$ ✓
- d. $12 + 3i$

$$z_1 + z_2 = (3 - 9) + (7 + 4)i = -6 + 11i.$$

21 Let $z_1 = 5 + 2i$ and $z_2 = 8 - 2i$. Find $z_1 + z_2$.

- a. 13 ✓
- b. $-3 + 4i$
- c. $13 - 2i$
- d. 15

$$z_1 + z_2 = (5 + 8) + (2 - 2)i = 13.$$

22 Let $z_1 = 9 - 7i$ and $z_2 = -4 - 8i$. Find $z_1 + z_2$.

- a. $7 - 15i$
- b. $5 - 15i$ ✓
- c. $13 + i$
- d. $5 - 17i$

$$z_1 + z_2 = (9 - 4) + (-7 - 8)i = 5 - 15i.$$

23 Let $z_1 = -8 - i$ and $z_2 = -5 + 6i$. Find $z_1 + z_2$.

- a. $-13 + 3i$
- b. $-3 - 7i$
- c. $-11 + 5i$
- d. $-13 + 5i$ ✓

$$z_1 + z_2 = (-8 - 5) + (-1 + 6)i = -13 + 5i.$$

24 Let $z_1 = -3 + 5i$ and $z_2 = 8 + 3i$. Find $z_1 + z_2$.

a. $7 + 8i$

b. $5 + 6i$

c. $-11 + 2i$

d. $5 + 8i$ ✓

$$z_1 + z_2 = (-3 + 8) + (5 + 3)i = 5 + 8i.$$

Multiple choice — medium

25 Solve $x^2 - 2x + 5 = 0$.

a. $1 \pm 2i$ ✓

b. $1 \pm 3i$

c. $2 \pm 2i$

d. $-1 \pm 2i$

$$\Delta = (-2)^2 - 4(1)(5) = 4 - 20 = -16 < 0. x = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i.$$

26 Solve $x^2 - 10x + 74 = 0$.

a. $5 \pm 8i$

b. $6 \pm 7i$

c. $-5 \pm 7i$

d. $5 \pm 7i$ ✓

$$\Delta = (-10)^2 - 4(1)(74) = 100 - 296 = -196 < 0. x = \frac{10 \pm \sqrt{-196}}{2} = \frac{10 \pm 14i}{2} = 5 \pm 7i.$$

27 Solve $x^2 + 2x + 65 = 0$.

- a. $-1 \pm 9i$
- b. $-1 \pm 8i$ ✓
- c. $1 \pm 8i$
- d. $0 \pm 8i$

$$\Delta = 2^2 - 4(1)(65) = 4 - 260 = -256 < 0. x = \frac{-2 \pm \sqrt{-256}}{2} = \frac{-2 \pm 16i}{2} = -1 \pm 8i.$$

28 Solve $x^2 + 12x + 72 = 0$.

- a. $6 \pm 6i$
- b. $-6 \pm 6i$ ✓
- c. $-6 \pm 7i$
- d. $-5 \pm 6i$

$$\Delta = 12^2 - 4(1)(72) = 144 - 288 = -144 < 0. x = \frac{-12 \pm \sqrt{-144}}{2} = \frac{-12 \pm 12i}{2} = -6 \pm 6i.$$

29 Solve $x^2 + 8x + 32 = 0$.

- a. $-4 \pm 4i$ ✓
- b. $-4 \pm 5i$
- c. $-3 \pm 4i$
- d. $4 \pm 4i$

$$\Delta = 8^2 - 4(1)(32) = 64 - 128 = -64 < 0. x = \frac{-8 \pm \sqrt{-64}}{2} = \frac{-8 \pm 8i}{2} = -4 \pm 4i.$$

30 Solve $x^2 - 6x + 58 = 0$.

- a. $3 \pm 7i$ ✓
- b. $4 \pm 7i$
- c. $-3 \pm 7i$
- d. $3 \pm 8i$

$$\Delta = (-6)^2 - 4(1)(58) = 36 - 232 = -196 < 0. x = \frac{6 \pm \sqrt{-196}}{2} = \frac{6 \pm 14i}{2} = 3 \pm 7i.$$

31 Solve $x^2 + 8x + 65 = 0$.

a. $4 \pm 7i$

b. $-4 \pm 7i$ ✓

c. $-3 \pm 7i$

d. $-4 \pm 8i$

$$\Delta = 8^2 - 4(1)(65) = 64 - 260 = -196 < 0. x = \frac{-8 \pm \sqrt{-196}}{2} = \frac{-8 \pm 14i}{2} = -4 \pm 7i.$$

32 Solve $x^2 - 10x + 41 = 0$.

a. $5 \pm 4i$ ✓

b. $-5 \pm 4i$

c. $5 \pm 5i$

d. $6 \pm 4i$

$$\Delta = (-10)^2 - 4(1)(41) = 100 - 164 = -64 < 0. x = \frac{10 \pm \sqrt{-64}}{2} = \frac{10 \pm 8i}{2} = 5 \pm 4i.$$

33 Solve $x^2 - 4x + 20 = 0$.

a. $3 \pm 4i$

b. $-2 \pm 4i$

c. $2 \pm 5i$

d. $2 \pm 4i$ ✓

$$\Delta = (-4)^2 - 4(1)(20) = 16 - 80 = -64 < 0. x = \frac{4 \pm \sqrt{-64}}{2} = \frac{4 \pm 8i}{2} = 2 \pm 4i.$$

34 Solve $x^2 - 14x + 98 = 0$.

- a. $8 \pm 7i$
- b. $7 \pm 8i$
- c. $-7 \pm 7i$
- d. $7 \pm 7i$ ✓

$$\Delta = (-14)^2 - 4(1)(98) = 196 - 392 = -196 < 0. x = \frac{14 \pm \sqrt{-196}}{2} = \frac{14 \pm 14i}{2} = 7 \pm 7i.$$

35 Solve $x^2 - 6x + 45 = 0$.

- a. $3 \pm 7i$
- b. $4 \pm 6i$
- c. $-3 \pm 6i$
- d. $3 \pm 6i$ ✓

$$\Delta = (-6)^2 - 4(1)(45) = 36 - 180 = -144 < 0. x = \frac{6 \pm \sqrt{-144}}{2} = \frac{6 \pm 12i}{2} = 3 \pm 6i.$$

36 Solve $x^2 - 12x + 85 = 0$.

- a. $6 \pm 7i$ ✓
- b. $-6 \pm 7i$
- c. $7 \pm 7i$
- d. $6 \pm 8i$

$$\Delta = (-12)^2 - 4(1)(85) = 144 - 340 = -196 < 0. x = \frac{12 \pm \sqrt{-196}}{2} = \frac{12 \pm 14i}{2} = 6 \pm 7i.$$

37 Solve $x^2 - 16x + 113 = 0$.

- a. $9 \pm 7i$
- b. $8 \pm 8i$
- c. $-8 \pm 7i$
- d. $8 \pm 7i$ ✓

$$\Delta = (-16)^2 - 4(1)(113) = 256 - 452 = -196 < 0. x = \frac{16 \pm \sqrt{-196}}{2} = \frac{16 \pm 14i}{2} = 8 \pm 7i.$$

38 Solve $x^2 - 14x + 74 = 0$.

a. $7 \pm 6i$

b. $7 \pm 5i$ ✓

c. $8 \pm 5i$

d. $-7 \pm 5i$

$$\Delta = (-14)^2 - 4(1)(74) = 196 - 296 = -100 < 0. x = \frac{14 \pm \sqrt{-100}}{2} = \frac{14 \pm 10i}{2} = 7 \pm 5i.$$

39 Solve $x^2 + 16x + 80 = 0$.

a. $8 \pm 4i$

b. $-7 \pm 4i$

c. $-8 \pm 4i$ ✓

d. $-8 \pm 5i$

$$\Delta = 16^2 - 4(1)(80) = 256 - 320 = -64 < 0. x = \frac{-16 \pm \sqrt{-64}}{2} = \frac{-16 \pm 8i}{2} = -8 \pm 4i.$$

Multiple choice — hard

40 Find $|z|$ for $z = -7 + 24i$.

a. 23

b. 25 ✓

c. 31

d. 27

$$|z| = \sqrt{(-7)^2 + 24^2} = \sqrt{625} = 25.$$

41 Find $|z|$ for $z = -1 + 6i$.

a. 37

b. $\sqrt{39}$

c. $\sqrt{35}$

d. $\sqrt{37}$ ✓

$$|z| = \sqrt{(-1)^2 + 6^2} = \sqrt{37}.$$

42 Find $|z|$ for $z = -1 - 4i$.

a. $\sqrt{17}$ ✓

b. $\sqrt{15}$

c. 17

d. $\sqrt{19}$

$$|z| = \sqrt{(-1)^2 + (-4)^2} = \sqrt{17}.$$

43 Find $|z|$ for $z = -5 + 12i$.

a. 15

b. 11

c. 13 ✓

d. 17

$$|z| = \sqrt{(-5)^2 + 12^2} = \sqrt{169} = 13.$$

44 Find $|z|$ for $z = -12 - 35i$.

a. 47

b. 39

c. 35

d. 37 ✓

$$|z| = \sqrt{(-12)^2 + (-35)^2} = \sqrt{1369} = 37.$$

45 Find $(2 - 5i) - (4 - 2i) + (1 - i)$.

a. $1 - 4i$

b. $-1 - 6i$

c. $2 - 4i$

d. $-1 - 4i$ ✓

$$(2 - 4 + 1) + (-5 + 2 - 1)i = -1 - 4i.$$

46 Find $(2 - 7i) - (-6 - 9i) + (-8 - 5i)$.

a. $0 - 5i$

b. $3 - 3i$

c. $0 - 3i$ ✓

d. $2 - 3i$

$$(2 + 6 - 8) + (-7 + 9 - 5)i = 0 - 3i.$$

47 Find $(-7 - 4i) - (-8 - 9i) + (-7 + 9i)$.

a. $-4 + 14i$

b. $-6 + 14i$ ✓

c. $-6 + 12i$

d. $6 + 14i$

$$(-7 + 8 - 7) + (-4 + 9 + 9)i = -6 + 14i.$$

48 Find $(8 + i) - (5 - 3i) + (-8 + 9i)$.

a. $-5 + 11i$

b. $-5 + 13i$ ✓

c. $-3 + 13i$

d. $5 + 13i$

$$(8 - 5 - 8) + (1 + 3 + 9)i = -5 + 13i.$$

49 Find $(-2 + 5i) - (6 + 6i) + (-6 + 6i)$.

a. $-14 + 5i$ ✓

b. $-14 + 3i$

c. $14 + 5i$

d. $-12 + 5i$

$$(-2 - 6 - 6) + (5 - 6 + 6)i = -14 + 5i.$$

50 Multiply $(8 + 4i)(1 + 3i)$.

a. $-4 + 28i$ ✓

b. $20 + 28i$

c. $-4 + 30i$

d. $-4 + 20i$

$$8 + 24i + 4i + 12i^2 = 8 + 28i + 12(-1) = -4 + 28i.$$

51 Multiply $(3 - i)(6 + 7i)$.

a. $25 + 27i$

b. $25 + 15i$ ✓

c. $25 + 17i$

d. $11 + 15i$

$$18 + 21i - 6i - 7i^2 = 18 + 15i - 7(-1) = 25 + 15i.$$

52 Multiply $(-5 + 2i)(-2 + i)$.

a. $8 - 7i$

b. $12 - 9i$

c. $8 - i$

d. $8 - 9i$ ✓

$$10 - 5i - 4i + 2i^2 = 10 - 9i + 2(-1) = 8 - 9i.$$

53 Multiply $(6 - 4i)(-9 + i)$.

a. $-50 + 44i$

b. $-50 - 30i$

c. $-50 + 42i$ ✓

d. $-58 + 42i$

$$-54 + 6i + 36i - 4i^2 = -54 + 42i - 4(-1) = -50 + 42i.$$

54 Multiply $(9 - 8i)(8 - 6i)$.

a. $120 - 118i$

b. $24 - 118i$ ✓

c. $24 - 116i$

d. $24 + 10i$

$$72 - 54i - 64i + 48i^2 = 72 - 118i + 48(-1) = 24 - 118i.$$